Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The equation of the stationary wave is
$y = 2A \sin\left(\frac{2\pi ct}{\lambda}\right) \cos\left(\frac{2\pi x}{\lambda}\right)$
Which of the following statements is wrong?

(A) The unit of ct is same as that of λ.

(B) The unit of x is same as that of λ.

(C) The unit of $2\pi c/\lambda$ is same as that of $2\pi x/\lambda t$.

(D) The unit of c/λ is same as that of x/λ.

Q.2

The dimensional representation of Planck's constant is identical to that of

(A) torque

(B) work

(C) stress

(D) angular momentum

Q.3

The dimensions of shear modulus of rigidity are

(A) $M^1L^1T^{-2}$

(B) $M^1L^1T^{-1}$

(C) ML^2T^2

(D) $ML^{-1}T^{-2}$

Q.4

Suppose refractive index μ is given as $\mu = A + \frac{B}{\lambda^2}$ where A and B are constants and λ is wavelength, then dimensions of B are same as that of

(A) wavelength

(B) volume

(C) pressure

(D) area

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Q.5

The frequency (n) of vibration of a string is given as $n = \frac{1}{2l}\sqrt{\frac{T}{m}}$, where T is tension and l is the length of vibrating string. Then, the dimensional formula for m is

(A) $\left[M^0L^1T'\right]$

(B) $\left[M^0L^0T^0\right]$

(C) $\left[M^1L^{-1}T^0\right]$

(D) $ML^0T^0\right]$

Q.6

Which of the following is not measured in units of energy?

(A) Couple $\times$ angle turned through

(B) Moment of inertia $\times$ (angular velocity) ^{2}

(C) Force $\times$ distance

(D) Impulse $\times$ time

Q.7

Dimensions of $\varepsilon_0\mu_0$ are

(A) $\left[LT^{-1}\right\}$

(B) $\left\{LT^{-2}\right]$

(C) $\left[L^2T^{-2}\right]$

(D) $\left[L^{\cdots-2}T^2\right]$

Q.8

If C, the velocity of light, g the acceleration due to gravity and P the atmosphenic pressure be the fundamental quantities in M.K.S. system, then the dimensions of length will be same as that of

(A) C/g

(B) C/P

(C) PCg

(D) C^2/g

Q.9

The frequency f of vibrations of a mass m suspended from a spring of spring consant k is given by $f = Cm^*k^y$, where C is a dimensionless constant. The values of x and y are, respectively

(A) $\frac{1}{2}, \frac{l}{2}$

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(B) $-\frac{1}{2}, -\frac{1}{2}$

(C) $\frac{1}{2}, -\frac{1}{2}$

(D) $-\frac{1}{2}, \frac{1}{2}$

Q.10

The number of particles crossing a unit area perpendicular to the X-axis in a unit time is given by $n = -D\left(\frac{n_2-n_1}{x_2-x_1}\right)$, where n_1 and n_2 are the number of particles per unit volume at $x = x_1$ and $x = x_2$, respectively, and D is the diffusion constant The dimensions of D are

(A) $\left[M^0LT^{-2}\right]$

(B) $\left[M^0L^2T^{-4}\right]$

(C) $\left[M^0L^2T^{-2}\right]$

(D) $\left[M^0L^2T^{-1}\right]$

Q.11

The moment of inertia of a body rotating about a given axis is 12.0kgm^2 in the S.I. system. What is the value of the moment of inertia in a system of units in which the units of length is $5\ \text{cm}$ and the unit of mass is $10\ \text{g}$?

(A) 2.4×10^3

(B) 6.0×10^3

(C) 5.4×10^5

(D) 4.8×10^5

Q.12

Which of the following product of e, h, μ, G (where μ is the permeability) be taken so that the dimensions of the product are same as that of speed of light?

(A) $he^{-2}\mu^{-1}G^0$

(B) $h^2eG^0\mu$

(C) $h^0e^2G^{-1}\mu$

(D) $hGe^{-2}\mu^0$

Q.13

Which of the following quantities has its unit as newtonsecond?

(A) Energy

(B) Torque

(C) Momentum

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(D) Angular momentum

Q.14

If force (F), acceleration (A) and time (T) be taken as the fundamental physical quantities, the dimensions of length on this systen of units are

(A) FAT^2

(B) FAT

(C) FT

(D) AT^2

Q.15

The dimensions of intensity of a wave are

(A) $\left[ML^2T^{-3}\right]$

(B) $\left[ML^0T^{-3}\right]$

(C) $\left[ML^{-2}T^{-3}\right]$

(D) $\left[M^1L^2T^3\right]$

Q.16

What are the dimensions of Gas constant?

(A) $\left[MLT^{-2}K^{-1}\right]$

(B) $\left[M^0LT^{-2}K^{-1}\right]$

(C) $\left[ML^2T^{\cdots 2}K^{-1}\ \mathrm{mol}^{-1}\right]$

(D) $\left[M^0L^2T^{-2}K^{\cdots}\right]$

Q.17

The dimensions of $\frac{1}{2}\varepsilon_0 E^2$ ($\varepsilon_0 =$ permittivity of free space and $E =$ electric field) are

(A) $\left| ML^2T^{-1}\right]$

(B) $\left[ML^{-1}T^{-2}\right]$

(C) $\left[ML^2T^{\cdots 2}\right]$

(D) $\left[MLT^{-1}\right]$

Q.18

Which of the following does not have the dimensions of velocity? (Given, $\varepsilon_0 =$ permittivity of free space; $\mu_0 =$ permittivity of free space; $v =$ frequency; $\lambda =$ wavelength; $P =$ pressure; $\rho =$ density; $k =$ wave number; and $\omega =$ angular frequency.

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(A) $\omega \dot{k}$

(B) $v\lambda$

(C) $\frac{1}{\sqrt{\varepsilon_0 \mu_0}}$

(D) $\sqrt{\frac{P}{\rho}}$

Q.19

In the formula $X = 3YZ^2$, X and Z have the dimensions of capacitance and magnetic induction, respectively. The dimensions of Y in M.K.S. system are

(A) $M^{-3}L^{-2}T^{-2}A^{-4}$

(B) $ML^{-2}\vec{A}$

(C) $M^{-3}L^{-2}T^{8}A^{4}$

(D) $M^{-3}L^{-2}T^{4}A^{4}$

Q.20

A length is measured as $7.60\ \mathrm{m}$, This is the same as

(A) $7600\ \mathrm{mm}$

(B) $0.0076\ \mathrm{mm}$

(C) $760\ \mathrm{cm}$

(D) $0.76\mathrm{dm}$

Q.21

The unit of surface tension may be expressed as

(A) joule metre

(B) newton metre

(C) joule metre $^{-2}$

(D) newton metre $^{-2}$

Q.22

Given that: $y = A\sin\left[\left(\frac{2\pi}{\lambda}\right)(ct - x)\right]$, where y and x are measured in the unit of length, Which of the following statements is true?

(A) The unit of λ is same as that of x and A.

(B) The unit of λ is same as that of x but not of A.

(C) The unit of c is same as that of $2\pi/\lambda$

(D) The unit of $(ct - x)$ is same as that of $2\pi/\lambda$

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Q.23

A physical quantity x depends on quantities y and z as follows: $x = Ay + B\tan(Cz)$, where A, B and C are constants. Which of the following do not have the same dimensions?

(A) x and B

(B) C and z^{-1}

(C) y and B/A

(D) x and A

Q.24

In the relation $y = r\sin(\omega t - kx)$, the dimensions of ω/k are

(A) $\left[M^0L^0T^0\right]$

(B) $\left[M^0L^1T^{-1}\right]$

(C) $\left[M^0L^0T^1\right]$

(D) $\left[M^0L'T^0\right]$

Q.25

If the velocity (V), acceleration (A) and force (F) are taken as fundamental quantities instead of mass (M), length (L) and time (T), the dimensions of Young's modulus (Y) would be

(A) FA^2V^{-4}

(B) FA^2V^{-5}

(C) FA^2V^{-3}

(D) FA^2V^{-2}

Q.26

If x and a stand for distance, then for what value of n is the given equation dimensionally correct? The equation is $\int \frac{dx}{\sqrt{a^2-x^n}} = \sin^{-1}\frac{x}{a}$.

Q.27

The number of significant figures in 5.69×10^{15} kg is

Q.28

The relative density of a material is found by weighing the body first in air and then in water. If the weight in air is (10.0 ± 0.1) gf and weight in water is (5.0 ± 0.1)gf, then the maximum permissible percentage error in relative density is

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Q.29

In the formula $X = 3YZ^2$, X and Z have the dimensions of capacitance and magnetic induction, respectively. The dimensions of Y in M.K.S system are $M^pL^qT^rA^5$, then the value of $p \times q \times r \times s$ is

Q.30

In the relation $P = \frac{\alpha}{\beta}e^{-\alpha z/K\theta}$ temperature. The dimensional formula of β will be $\left[M^xL^yT^z\right]$, then the value of $x + y + z$ is

Answers & Solutions

Q.1 Answer:

The unit of c/λ is same as that of x/λ.

Solution:

Here, $(2\pi ct/\lambda)$ as well as $(2\pi x/\lambda)$ are dimensionless.
So, unit of ct is same as that of λ.
Unit of x is same as that of λ.
Also, $\left[\frac{2\pi ct}{\lambda}\right] = \left[\frac{2\pi x}{\lambda}\right] = M^0L^0T^0$
Hence, $\left[\frac{2\pi c}{\lambda}\right] = \left[\frac{2\pi x}{\lambda t}\right]$.
In the option (d), $\frac{x}{\lambda}$ is unit less.
This is not the case with c/λ.

Q.2 Answer:

angular momentum

Solution:

Angular momentum, $J = mvr$
$[J] = \left[MLT^{-1}L\right] = \left[ML^2T^{-1}\right]$
This is same as that of Planck's constant.

Q.3 Answer:

$ML^{-1}T^{-2}$

Solution:

$$\eta = \frac{\text{tangential stress}}{\text{shearing strain}} = \frac{T}{\theta} = \frac{F/A}{x/L}$$
$$\Rightarrow \quad [\eta] = \frac{MLT^{-2}}{L^2} = \left[ML^{-1}T^{-2}\right]$$

Q.4 Answer:

area

Solution:

As $\mu = \frac{\text{velocity of light in vaceum}}{\text{velocity of light in medium}}$

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hence, μ is dimensionless.

Thus, each term on the R.H.S. of given equation should be dimensionless, i.e., $\frac{B}{\lambda^2}$ is dimensionless, i.e., B should have dimension of λ^2, i.e., cm^2, i.e., area.

Q.5 Answer:

$\left[M^1L^{-1}T^0\right]$

Solution:

$n=\frac{1}{2l}\sqrt{\frac{T}{m}} \Rightarrow n^2=\frac{1}{4l^2}\frac{T}{m}$

$n=\frac{T}{4l^2n^2}=\left[\frac{MLT^{-2}}{L^2\times T^{-2}}\right]=\left[ML^{-1}\right]$

Q.6 Answer:

Impulse $\times$ time

Solution:

$\text{Energy } = M^1L^2T^{-2}$

Couple $\times$ (angle turned through) $=\overrightarrow{\mathrm{r}}\times\overrightarrow{\mathrm{f}}\times\theta=\left(\mathrm{L}^1\right)\left(\mathrm{M}^1\ \mathrm{L}^1\ \mathrm{T}^{-2}\right)\left(\mathrm{M}^{\circ}\mathrm{L}^{\circ}\mathrm{T}^0\right)=\mathrm{M}^1\ \mathrm{L}^2\ \mathrm{T}^{-2}$

(Moment of Inertia) $\times$ (Angular velocity) $^2=\left(\mathrm{MR}^2\right)\times\left(\omega^2\right)=$

$\left(\mathrm{M}^1\right)\left(\mathrm{L}^2\right)\left(\mathrm{T}^{-2}\right)=\mathrm{M}^1\ \mathrm{L}^2\ \mathrm{T}^{-2}$

Force $\times$ distance $=\mathrm{M}^1\ \mathrm{L}^1\ \mathrm{T}^{-2}\times\mathrm{L}^1=\mathrm{ML}^2\ \mathrm{T}^{-2}$

Impulse $\times$ time $=(\mathrm{f}\times\mathrm{t})\times\mathrm{t}=\mathrm{M}^1\ \mathrm{L}^1\ \mathrm{T}^{-2}\times\mathrm{T}\times\mathrm{T}=\mathrm{M}^1\ \mathrm{L}^1\rightarrow$ **not a unit of energy**

Q.7 Answer:

$\left[L^{\cdots-2}T^2\right]$

Solution:

We know that speed of e.m. wave is

$C=\frac{1}{\sqrt{\mu_0\varepsilon_0}}\Rightarrow\mu_0\varepsilon_0=\frac{1}{C^2}=\left[L^{-2}T^2\right]$

Q.8 Answer:

C^2/g

Solution:

$\frac{C^2}{g}=\frac{L^2T^{-2}}{LT^{-2}}=[L]$

Q.9 Answer:

$-\frac{1}{2},\frac{1}{2}$

Solution:

$f=Cm^xk^3$**. Writing dimensions on both sides:**

$\left[M^0L^0T^{-1}\right]=M^x\left[ML^0T^{-2}\right]^y$

$\left[M^0L^0T^{-1}\right]=\left[M^{x+y}T^{-2y}\right]$

Comparing dimensions on both sides, we have

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$0 = x + y$ and $-1 = -2y \Rightarrow y = \frac{1}{2}, x = -\frac{1}{2}$

Aliter. Remembering that frequency of oscillation of loaded spring is

$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}(k)^{1/2}m^{-1/2}$

which gives $x = -\frac{1}{2}$ **and** $y = \frac{1}{2}$

Q.10 Answer:

$\left[M^0L^2T^{-1}\right]$

Solution:

$n = -\frac{D(n_2-n_1)}{x_2-x_1} \Rightarrow T^{-1}L^{-2} = \frac{D\left(L^{-3}\right)}{L}$

$\Rightarrow \quad D = \frac{T^{-1}L^{-2}\times L}{L^{-3}} \Rightarrow D = \left[M^0L^2T^{-1}\right]$

Q.11 Answer:

4.8×10^5

Solution:

$n_2 = n_1\left(\frac{M_1}{M_2}\right)^a\left(\frac{L_1}{L_2}\right)^b\left(\frac{T_1}{T_2}\right)^0$

Dimensional formula of moment of inertia $= \left[ML^2T^0\right]$

$\therefore a = 1, b = 2, c = 0$

$n_1 = 12.0, M_1 = 1\text{ kg}, M_2 = 10\text{ g}$

$L_1 = 1\text{ m}_i L_2 = 5\text{ cm}, T_1 = 1\text{ s}, T_2 = 1\text{ s}$

$n_2 = 12.0\left(\frac{1\text{ kg}}{10\text{ g}}\right)^1\left(\frac{1\text{ m}}{5\text{ cm}}\right)^2\left(\frac{1\text{sec}}{1\text{sec}}\right)^0$

$= 12 \times \left(\frac{1000\text{ g}}{10\text{ g}}\right)^1\left(\frac{100\text{ cm}}{5\text{ cm}}\right)^2 \times 1$

$= 12 \times 100 \times 400 = 4.8 \times 10^5$

Q.12 Answer:

$he^{-2}\mu^{-1}G^0$

Solution:

Here $v = e^a h^b \mu^c G^d$**. Taking the dimensions**

$M^0LT^{-1}A^0 = \left[AT^l\right]^a\left[ML^2T^{-1}\right]^b\left[MLT^{-2}A^{-2}\right]^c$

$\left[M^{-1}L^3T^{-2}\right]^d$

There will be 4 simultaneous equations by equating the dimensions of M, L, T **and** A**. These are** $a - 2c = 0$

$a - b - 2c - 2d = -1, b + c - d = 0$

and $2b + c + 3d = 1$

Solving for ' a **',**b'**, '** c **' and '** d **' we get**

$a = -2, b = 1, c = -1, d = 0$

Thus, $v = e^{-2}h\mu^{-1}G^0$

Q.13 Answer:

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Momentum

Solution:

Momentum = force $\times$ time = Ns

Q.14 Answer:

AT^2

Solution:

$$AT^2 = LT^{-2} \times T^2 = \left[M^0LT^0\right]$$

Q.15 Answer:

$\left[ML^0T^{-3}\right]$

Solution:

Intensity of wave

$$= \frac{\text{energy}}{\text{area} \times \text{time}} = \frac{ML^2T^{-2}}{L^2T} = \left[ML^0T^{-3}\right]$$

Q.16 Answer:

$\left[ML^2T^{\cdots 2}K^{-1}\ \text{mol}^{-1}\right]$

Solution:

$$PV = nRT \Rightarrow R = \frac{PV}{nT} = \frac{ML^{-1}T^{-2} \times L^3}{mol \times K}$$
$$= \left[ML^2T^{-2}K^{-1}\ \text{mol}^{-1}\right]$$

Q.17 Answer:

$\left[ML^{-1}T^{-2}\right]$

Solution:

$\frac{1}{2}\varepsilon_0 E^2$ is the expression for electrostatic energy density, i.e., the energy stored per unit volume in a parallel plate capacitor.

$$\therefore \frac{1}{2}\varepsilon_0 E^2 = \frac{\text{energy}}{\text{volume}}$$
$$= \frac{ML^2T^{-2}}{L^3} = \left[ML^{-1}T^{-2}\right]$$

Alternatively, $\varepsilon_0 = \frac{1}{4\pi F} \times \frac{q_1 q_2}{r^2}$ **and** $E = \frac{F}{q}$

$\therefore \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{8\pi F}\frac{q1q2}{r^2} \times \frac{F^2}{q^2} = \frac{F}{r^2} = \frac{MLT^{-2}}{L^2}$

Q.18 Answer:

$\omega \dot{k}$

Solution:

Here, $\omega \times k = T^{-1}L^{-1}$ which are not the dimensions of velocity. All others have got the dimensions of velocity.

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Q.19 Answer:

$M^{-3}L^{-2}T^{8}A^{4}$

Solution:

Here, $X = \frac{Q}{V}$**. But** $V = \frac{W}{Q}$

$\therefore X = \frac{Q^2}{W}$**. Now,** $Z = B = \frac{F}{IL}$

$\therefore \quad Y = \frac{X}{3Z^2} = \frac{1}{3}\frac{Q^2}{W} \times \frac{I^2L^2}{F^2}$

$= \frac{A^2T^2A^2L^2}{[ML^2T^{-2}][M^2L^2T^{-4}]} = \left[M^{-3}L^{-2}T^{8}A^{4}\right]$

Q.20 Answer:

760 cm

Solution:

All the choices are equivalent but the answer must possess three significant digits as significant digits do not change on conversion from one system to another. So, appropriate choice is (c).

Q.21 Answer:

joule metre $^{-2}$

Solution:

Unit of surface tension is Nm^{-1}**.**

Also, $\mathrm{Im}^{-2} = \mathrm{Nmm}^{-2} = \mathrm{Nm}^{-1}$

Q.22 Answer:

The unit of λ is same as that of x and A.

Solution:

Here, unit of y **and** A **will be same and that of** x **and** λ **will be same.** $\frac{2\pi}{\lambda}(ct - x)$ **is dimensionless. Here,** $\frac{ct}{\lambda}$ **and** $\frac{x}{\lambda}$ **are dimensionless. Unit of** ct **is same as that of** λ **or** x**.**

Unit of ct must be same as unit of x**. Since argument of sin has to be dimensionless, thus dimensions of** x **and ct must be same as dimensions of** λ**. Thus dimension of** λ **equals '** L **'. Also, dimension of** y **equals '** L **' and hence dimension of** A **equals '** L **' because** $\sin$ **term is dimensionless. Thus statement 1 is correct.**

Q.23 Answer:

x and A

Solution:

$[x] = [Ay] = [B] \Rightarrow [y] = [B/A]$

Also, $[x] \neq [A]$ **and** $[Cz] =$ **dimensionless**

$\Rightarrow [C] = \left[z^{-!}\right]$

Q.24 Answer:

$\left[M^{0}L^{1}T^{-1}\right]$

Solution:

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b. $y = r\sin(\omega t - kx)$

Here, $\omega t =$ angle $\Rightarrow \quad \omega = \frac{1}{T} = T^{-1}$

Similarly, $kx =$ angle $\Rightarrow \quad k = \frac{1}{x} = L^{-1}$

$\therefore \quad \frac{\omega}{k} = \frac{T^{-1}}{L^{-1}} = LT^{-1}$

Or simply $\frac{\omega}{k}$ represents wave velocity $\frac{\omega}{k} = \frac{2\pi f}{2\pi/\lambda} = f\lambda = v$, where f is frequency

Q.25 Answer:

FA^2V^{-4}

Solution:

Let $Y = \left[V^a A^b F^c\right]$

$\left[ML^{-1}T^{-2}\right] = \left[LT^{-1}\right]^n \left[LT^{-2}\right]^b \left[MLT^{-2}\right]^c$

$ML^{\cdots-1}T^{\cdots2} = M^c L^{a+b+c} T^{-a-2b-2c}$

$\therefore c = 1, a + b + c = -1, -a - 2b - 2c = -2$

On solving, we get $a = -4, b = 2$ **and** $c = 1$

Q.26 Answer:

—

Solution:

Here, x^n **has the same dimensions as** a^2**. Thus,** $n = 2$ **will make the expression dimensionally correct.**

Q.27 Answer:

—

Solution:

5.69×10^{15} kg **has 3 significant figures as the power of 10 are not considered for significant figures.**

Q.28 Answer:

—

Solution:

Relative density $= \frac{W_a}{W_a - W_w}, \rho = \frac{W_a}{w}$

$\frac{\Delta\rho}{\rho} = \frac{\Delta W_a}{W_a} - \frac{\Delta w}{w}$

For maximum error, $\frac{\Delta\rho}{\rho} = \frac{\Delta W_a}{W_a} + \frac{\Delta w}{w}$

For maximum percentage error,

$\frac{\Delta\rho}{\rho} \times 100 = \frac{\Delta W_a}{W_a} \times 100 + \frac{\Delta w}{w} \times 100$

Given, $\Delta W_a = 0.1\text{gf}$ **and** $W_a = 10.0\text{gf}$

$w = 10.0 - 5.0 = 5.0\text{gf}$

$\Delta w = \Delta W_a + \Delta W_w = 0.1 + 0.1 = 0.2\text{gf}$

$\frac{\Delta\rho}{\rho} \times 100 = \left(\frac{0.1}{10.0}\right) \times 100 + \left(\frac{0.2}{5.0}\right) \times 100$

$= 1 + 4 = 5$

Q.29 Answer:

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Solution:

Here, $X = \frac{Q}{V}$. But $V = \frac{W}{Q}$

$\therefore \quad X = \frac{Q^2}{W}$. Now,$Z = B = \frac{F}{IL}$

$$Y = \frac{X}{3Z^2} = \frac{1}{3}\frac{Q^2}{W} \times \frac{I^2L^2}{F^2}$$

$$= \frac{A^2T^2A^2L^2}{\left[ML^2T^{-2}\right]\left[M^2L^2T^{-4}\right]} = \left[M^{-3}L^{-2}T^8A^4\right]$$

So, we get the value of $\mathrm{p} = -3, \mathrm{q} = -2, \mathrm{r} = 8,\ \mathrm{s} = 4$

$\therefore p \times q \times r \times s = 192$

Q.30 Answer:

—

Solution:

They've given the formula for Pressure as

$P = \frac{\alpha}{\beta}\exp\left(-\frac{\alpha z}{k\theta}\right)$

Where,

α, β are constants

z is distance

k is Boltzmann's constant

θ is temperature

Now, here we have to observe that the exponential does not have any units i.e.,

$\frac{\alpha z}{k\theta} = \left[M^0L^0T^0\right]$

We have the individual dimensional formulas for the quantities as

$z = \left[M^0L^1T^0\right]$

$k = \left[M^1L^2T^{-2}K^{-1}\right]$

$\theta = \left[M^0L^0T^0K^1\right]$

Substituting these quantities in the above formula,

$$\frac{\alpha z}{k\theta} = \left[M^0L^0T^0\right]$$

$$\Rightarrow \frac{\alpha\left[M^0L^1T^0\right]}{\left[M^1L^2T^{-2}K^{-1}\right]\left[M^0L^0T^0K^1\right]} = \left[M^0L^0T^0\right]$$

$$\Rightarrow \frac{\alpha\left[M^0L^1T^0\right]}{\left[M^1L^2T^{-2}K^0\right]} = \left[M^0L^0T^0\right]$$

$$\Rightarrow \alpha\left[M^{-1}L^{-1}T^2\right]$$

Now, writing the dimension formula for the pressure we have,

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$$P = \frac{\alpha}{\beta} \exp\left(-\frac{\alpha z}{k\theta}\right)$$

$$\Rightarrow [P] = \frac{[\alpha]}{[\beta]}$$

$$\Rightarrow \left[M^1 L^{-1} T^{-2}\right] = \frac{\left[M^1 L^1 T^{-2}\right]}{[\beta]}$$

$$\Rightarrow [\beta] = \frac{\left[M^1 L^1 T^{-2}\right]}{\left[M^1 L^{-1} T^{-2}\right]}$$

$$\Rightarrow [\beta] = \left[M^0 L^2 T^0\right]$$

$$\therefore [\beta] = \left[L^2\right]$$

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